

AD7190

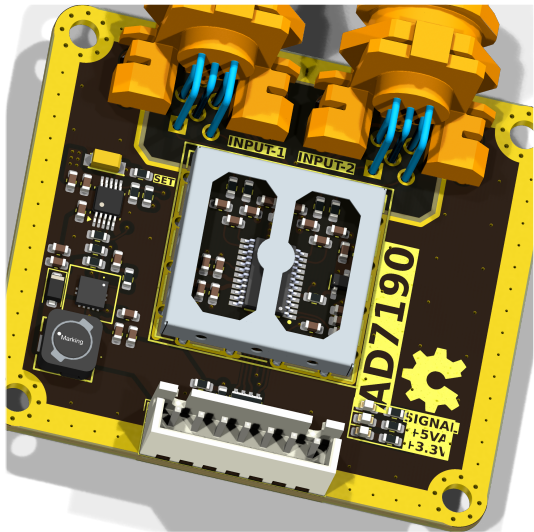
Variant:

2026-10-09

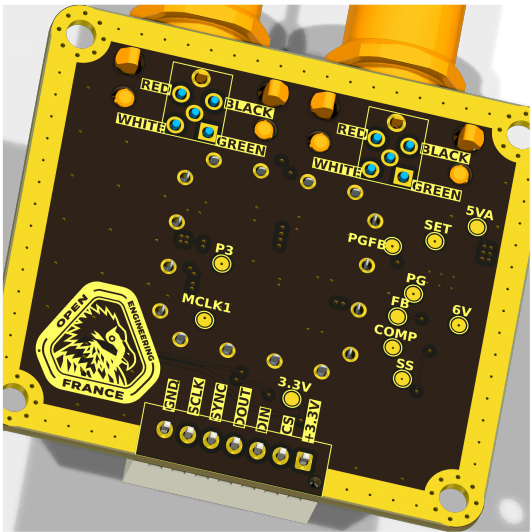
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

OPEN_WEIGHT - AD7190 24-bit load-cell front-end, part of OPEN THRUST LAB.
Single +3.3 V host supply, SPI interface, 2 x 4-wire bridge inputs.

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 09-Oct-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:

+3.3V host -> TPS61086
6 V -> LT3042 +5VA
(AVDD + excitation)

DEBUG NOTE:

TP on every rail.
Wait >= 600 ms (PG)
before ADC init.

DESIGN NOTE:

SPI mode 3,
SCLK <= 5 MHz,
RDY on DOUT.

DESIGN NOTE:

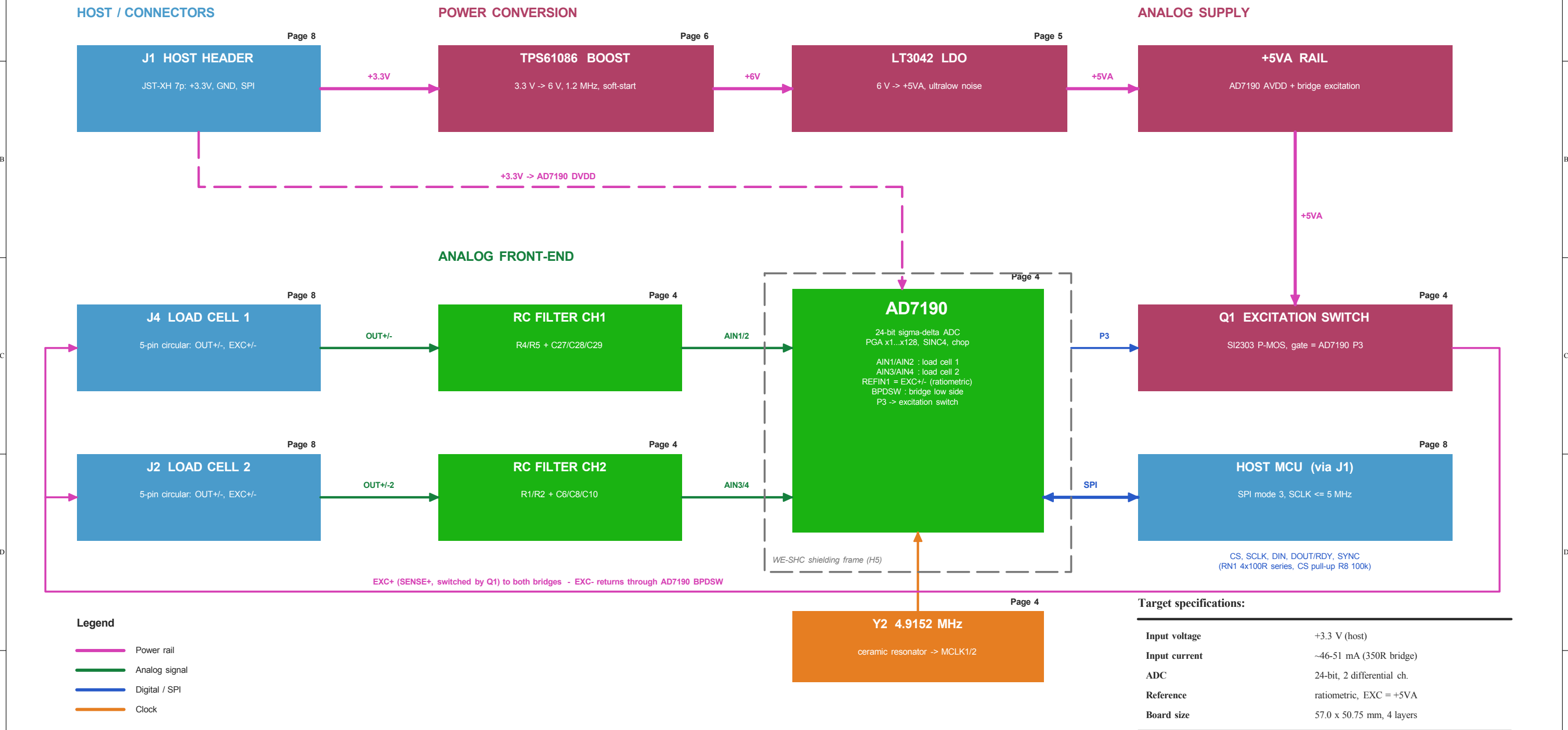
REFIN1 = bridge
excitation:
ratiometric.

LAYOUT NOTE:

AFE under the
WE-SHC shield H5,
boost SW node
away.

	Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
	Title: OPEN_WEIGHT_AD7190	Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Path: /	File Name: OPEN_WEIGHT.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
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[2] Block Diagram

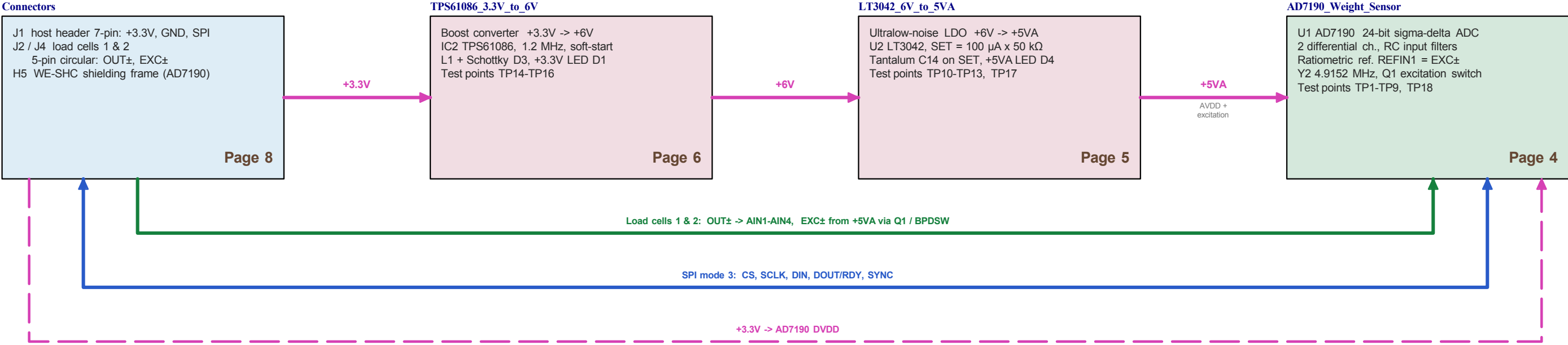


	Comments:		Company:		Variant:		Git Hash:		
			OPEN THRUST LAB				b3867f9		
			Board Name:			Project Name:			
			AD7190			OPEN_WEIGHT			
Sheet Title:		File Name:		Designer:		Date:		Revision:	
Block Diagram		Block Diagram.kicad_sch		SIEBERT DIMITRY		2026-09-29		+ (Unreleased)	
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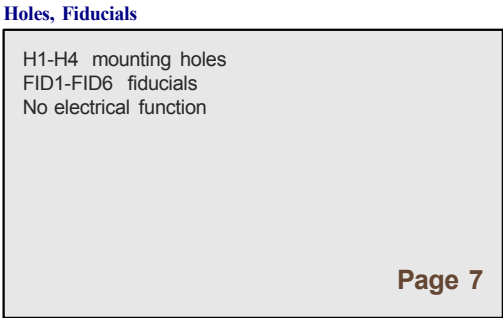
[3] Project Architecture

Sheet hierarchy and signal flow. Click a block or its page number to open the sheet.

HOST INTERFACE



MECHANICAL



LEGEND

- Power rail
- Secondary supply
- Analog signal
- Digital / SPI

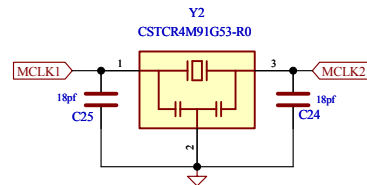
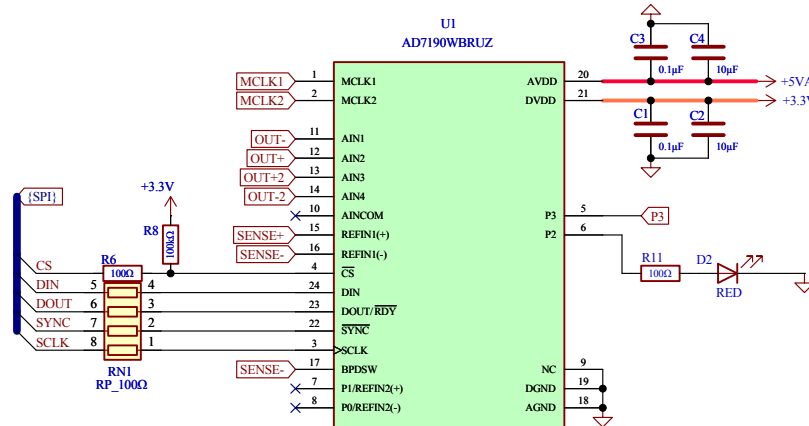
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- [7] Holes, Fiducials
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Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 10

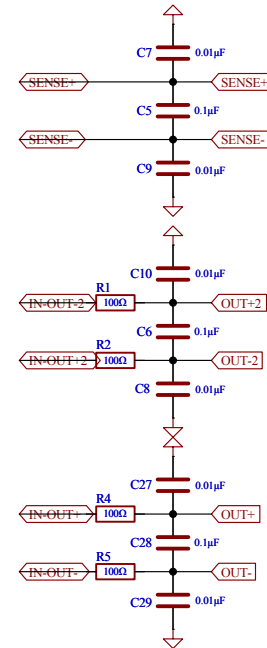
[4] AD7190 - Weight Sensor



**ADC FOR
weighing sensor**

DESIGN NOTE:

Ref Condo for quartz
Murata GRM1885C1H220JA01D
(22 pF, 50 V, C0G, 0603, ±5%)



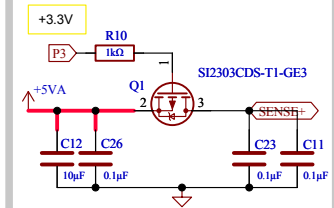
Calcul de filtre : $f_c = 1 / (2\pi \times R \times C) = X \text{ Hz}$

DESIGN NOTE:

Ref Condo for Filtre
Murata GRM1885C1H103JA01D (10 nF, 50 V, C0G, 0603, ±5%)
KEMET C0603R104K5RACTU (100 nF, 50 V, X7R, 0603, ±10%)

Ref Resistor for Filtre
TNPW0603100RBEEA (100 Ω, 0.1 W, thin film, 0603, ±0.1%, 25 ppm/°C)

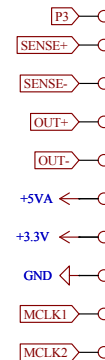
FILTRE



DESIGN NOTE:

IC1 is a P-MOS high-side switch used to enable/disable the load cell excitation. must disabled during calibration

DRIVE GATE



TEST POINT

About

The AD7190 is a low noise, complete analog front end for high precision measurement applications. It contains a low noise, 24-bit sigma-delta ($\Sigma\Delta$) analog-to-digital converter (ADC). The on-chip low noise gain stage means that signals of small amplitude can be interfaced directly to the ADC. The device can be configured to have two differential inputs or four pseudo differential inputs. The on-chip channel sequencer allows several channels to be enabled, and the AD7190 sequentially converts on each enabled channel. This simplifies communication with the part. The on-chip 4.92 MHz clock can be used as the clock source to the ADC or, alternatively, an external clock or crystal can be used. The output data rate from the part can be varied from 4.7 Hz to 4.8 kHz. The device has two digital filter options. The choice of filter affects the rms noise/noise-free resolution at the programmed output data rate, the settling time, and the 50 Hz/60 Hz rejection.



Comments:

AD7190
CSTCR4M91G53-R0
AD7190_TEST_BOARD CN

Company:

OPEN THRUST LAB

Board Name:

AD7190

Sheet Title:

AD7190 - Weight Sensor

Sheet Path:

/Project Architecture/AD7190_Weight_Sensor/

File Name:

AD7190 - Weight Sensor.kicad_sch

Designer:

SIEBERT DIMITRY

Reviewer:

Variant:

Git Hash:

b3867f9

Project Name:

OPEN_WEIGHT

Date:

2026-09-29

Revision:

+(Unreleased)

Size:

A4

Sheet:

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[6] TPS61086 - Boost 3.3V to 6V

COMP (pin 1): error amplifier compensation pin; connect the Rcomp + Ccomp network to AGND to stabilize the loop (do not route it near SW).

FB (pin 2): feedback input; connect it to the R1/R2 resistor divider from VOUT (after the diode) to set the output voltage (e.g. 6 V).

EN (pin 3): enable / shutdown; high level = active, low level = off. Often tied to IN for "always ON", or driven by a GPIO.

AGND (pin 4): analog ground (reference for sensitive signals: FB/COMP/SS/MODE). Keep it "clean" and connect it to the GND plane in a controlled way.

PGND (pin 5): power ground (return path for switching currents). Connect it to the GND plane with a short/wide copper area close to the power capacitors.

SW (pins 6-7): switch node; connect it to L and the Schottky diode. This is the noisiest node → keep the trace short, minimize the loop area, and keep it away from FB/COMP.

IN (pin 8): power input (2.3-6 V); decouple it as close as possible with Cby ~1 μF + Cin (e.g. 10 μF) to GND.

MODE (pin 9): mode selection; MODE = High → Forced PWM (more stable frequency, better for noise), MODE = Low → PFM (better efficiency at light load).

SS (pin 10): soft-start; add a capacitor (C_{SS}) to limit inrush current at startup. Left open = no soft-start.

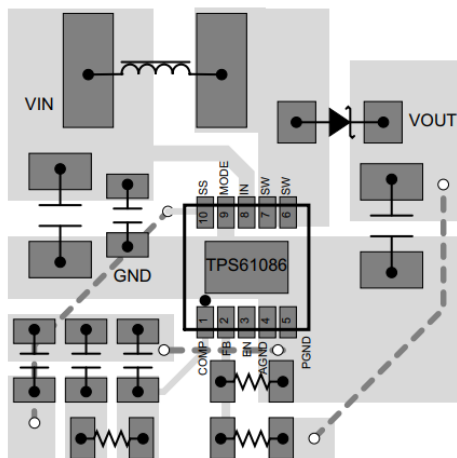
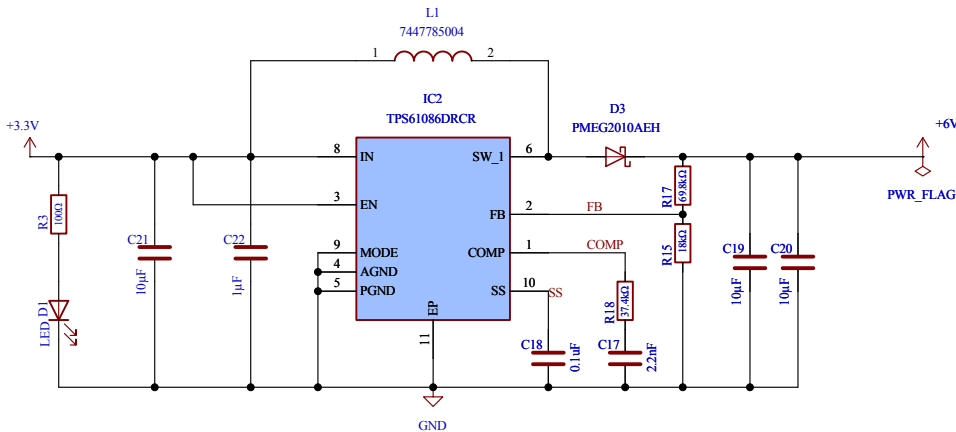


Figure 26. TPS61086 Layout Example

Pin usage and layout example



Boost 3.3 V -> 6.04 V

About

The TPS61086 device is a high-frequency, high-efficiency DC-to-DC converter with an integrated 2.0-A, 0.13-Ω power switch capable of providing an output voltage up to 18.5 V. The implemented boost converter is based on a fixed frequency of 1.2-MHz, pulse-width-modulation (PWM) controller that allows the use of small external inductors and capacitors and provides fast transient response.

At light-load, the device can operate in Power Save Mode with pulse-frequency-modulation (PFM) to improve the efficiency while keeping a low-output voltage ripple. For very noise-sensitive applications, the device can be forced to PWM Mode operation over the entire load range by pulling the MODE pin high. The external compensation allows optimizing the application for specific conditions. A capacitor connected to the soft-start pin minimizes inrush current at start-up.

Test points



Comments:		Company:		Variant:	Git Hash:
References:		OPEN THRUST LAB			b3867f9
TI TPS61086 datasheet L1 7447785004, D3 PMEG2010AEH		Board Name:		Project Name:	
		AD7190		OPEN_WEIGHT	
Sheet Title:	File Name:	Designer:	Date:	Revision:	
TPS61086 - Boost 3.3V to 6V	TPS61086 - Boost 3V3 to 6V.kicad_sch	SIEBERT DIMITRY	2026-09-29	+ (Unreleased)	
Sheet Path:	Reviewer:	Size:	Sheet:		
/Project Architecture/TPS61086_3.3V_to_6V/		A3	6 of 10		

[7] Holes, Fiducials										
H1 H2 H3 H4 GND HOLES FID1 Fiducial FID2 Fiducial FID3 Fiducial TOP FID4 Fiducial FID5 Fiducial FID6 Fiducial BOT										
					Comments: References: 4 x M3 (3.2 mm) plated holes to GND FID1-FID6 not fitted on this revision		Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
Sheet Title: Holes, Fiducials			File Name: Holes - Fiducials.kicad_sch		Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)		Project Name: OPEN_WEIGHT	
Sheet Path: /Project Architecture/Holes, Fiducials/					Reviewer:	Size: A3	Sheet: 7 of 10			



Comments: References: 4 x M3 (3.2 mm) plated holes to GND FID1-FID6 not fitted on this revision	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Title: Holes, Fiducials	File Name: Holes - Fiducials.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29
Sheet Path: /Project Architecture/Holes, Fiducials/	Reviewer:		Size: A3	Revision: + (Unreleased)
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[8] Connectors

DESIGN NOTE:
J1 host header (JST-XH, 7 pins):
1 +3.3V 2 CS 3 DIN 4 DOUT/RDY
5 SYNC 6 SCLK 7 GND

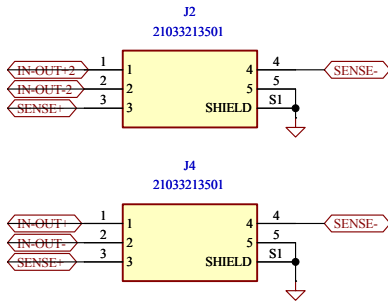
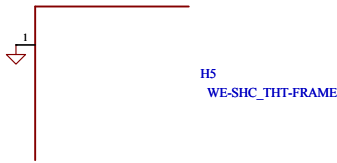
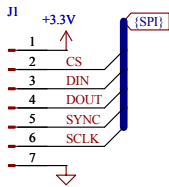
DESIGN NOTE:
J2/J4 load cell (5 pins):
1 OUT, 2 OUT (signal pair), 3 EXC+ (SENSE+),
4 EXC- (SENSE-/BPDSW), 5 GND/shield.
Both channels read V(pin2) - V(pin1):
fix the sign in firmware if needed.

LAYOUT NOTE:
H5 shielding frame covers the AD7190
front-end; connector shells tied to GND.

Pinout and notes



Power flags: +3.3V and GND come from the host connector J1,
(J2/J4 21033213501, H5 WE-SHC).



Host and load-cell connectors



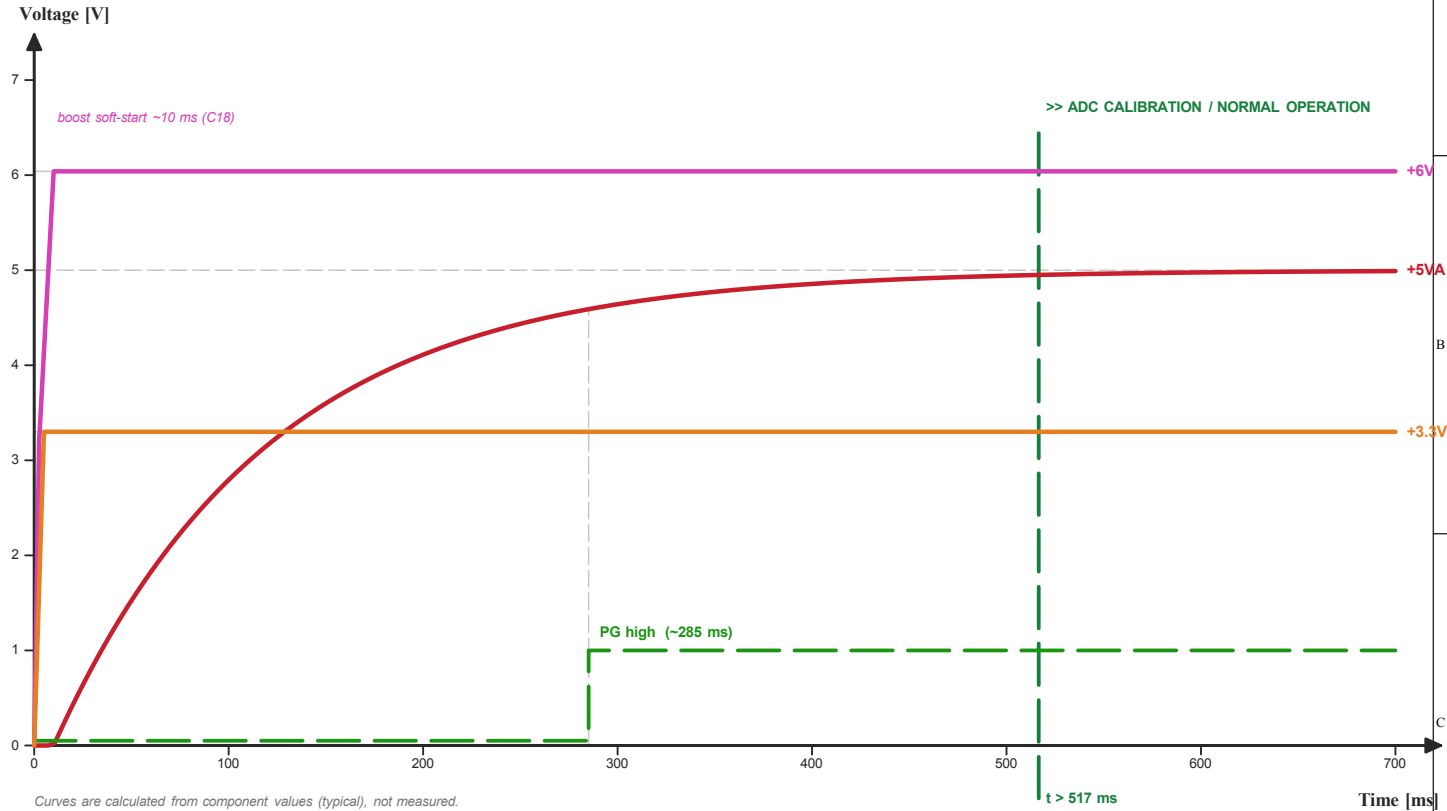
Comments: References: JST XH B7B-XH-AM (J1) J2/J4 21033213501, H5 WE-SHC	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
Sheet Title: Connectors	File Name: Connectors.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Connectors/		Reviewer:	Size: A3	Sheet: 8 of 10

[9] Power - Sequencing

NAME	SOURCE	LEVEL
+3.3V	Host (J1)	3.3 V (host)
+6V	TPS61086 IC2	6.04 V
+5VA	LT3042 U2	5.00 V
PG	LT3042 U2	> 4.59 V

Values from the schematic:
+6V = 1.238 V x (1 + R17/R15) = 1.238 x (1 + 69.8k/18k)
+5VA = ISET x RSET = 100 uA x 50k (R9)
PG threshold = 0.3 V x (1 + R12/R13) = 4.59 V
LT3042 soft-start: tau = RSET x CSET = 50k x 2.2 uF
= 110 ms -> +5VA at 99 % after ~550 ms

DESIGN NOTE:
The LT3042 SET capacitor (C14 = 2.2 uF) gives a 110 ms time constant. Firmware must wait >= 600 ms after power-up (or until PG, TP17) before the AD7190 reset and internal calibration. To start faster, reduce C14 (e.g. 0.47 uF -> tau = 23.5 ms) at the cost of slightly more noise.



	Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
		Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
	Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 9 of 10

[10] Revision History

Version Unreleased

Added

- AD7190 front-end, 2 load-cell inputs
- +3.3V -> TPS61086 6 V -> LT3042 +5VA
- Self-contained libs and 3D models (lib/)
- JLCPCB 4-layer rules and stackup
- LTspice simulations (Simulation/LTspice)
- openEMS layout coupling (Simulation/openEMS)

Changed

- Renamed to OPEN_WEIGHT, KiCad 10

Fixed

- Connectors not linked to ADC in schematic
- AD7190 solder-mask bridges

	Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
		Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Title: Revision History	File Name: Revision_History.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 10 of 10